Prabhavit Gupta

Agentic AI Engineer | Autonomous Code Pipelines, RAG Architectures & Claude Code Workflows

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Executive Summary

- **Autonomous Agentic Systems & AI Code Pipelines:** Agentic AI Engineer with 3 years commercial software experience architecting, shipping, and hardening production-grade autonomous agent loops, code-generation pipelines, and internal knowledge systems.
- **Deep Python, RAG & Vector Engineering:** Strong hands-on command of asynchronous, typed Python (FastAPI, asyncio, Pydantic), designing optimized PostgreSQL/pgvector indexing, multi-stage retrieval pipelines (embeddings, hybrid search, reciprocal rank fusion, re-ranking), and evaluation guardrails.
- **Agentic Development Mastery (Claude Code, MCP, LangGraph):** Daily power user and builder of agentic workflows with Claude Code, Cursor, and Model Context Protocol (MCP), creating automated test-generation/execution harnesses, Git/CI integrations, and continuous telemetry pipelines.

Technical Skills & Core Competencies

- **Agentic Frameworks & Tooling:** Model Context Protocol (MCP), LangGraph, Claude Code Development Workflows, Multi-Agent Tool Orchestration, Function Calling, Human-in-the-Loop Interrupts.
- **Python & Backend Engineering:** Python 3.11+ (Asyncio, Typing, Packaging, Concurrency), FastAPI, Pydantic V2, Celery, Redis Task Queues, RESTful APIs, Webhooks.
- **RAG & Database Architecture:** PostgreSQL, pgvector, Vector Indexing (HNSW, IVFFlat), Qdrant, ChromaDB, Cohere Re-rankers, Semantic Chunking, Reciprocal Rank Fusion (RRF).
- **Evaluation, Guardrails & Observability:** DeepEval, RAGAS, LangSmith, OpenTelemetry, Prometheus, Automated Test Generation (pytest), Input/Output Guardrails, Hallucination Mitigation.
- **DevOps, CI/CD & Infrastructure:** Docker, GitHub Actions CI/CD, Git-Based Workflows, Linux / Bash, Google Cloud Platform (Cloud Run, Artifact Registry), AWS.

Professional Experience

Agentic AI Engineer & Consultant • *Self-Employed / Independent AI Engineering Consultancy*
Oct 2024 – Present • Remote / Hybrid

- Architected and deployed autonomous multi-agent code generation and review systems using Claude Code, LangGraph, and custom Model Context Protocol (MCP) servers, accelerating feature delivery cycles across client squads.
- Built end-to-end RAG platforms over massive internal codebases and technical documentation, integrating dense vector embeddings, BM25 sparse retrieval, and Cohere re-ranking with sub-300ms query latency.
- Designed and optimized PostgreSQL schemas with pgvector HNSW indexing, query planning, and connection pooling to sustain high-concurrency vector similarity searches under production traffic.
- Engineered automated test-generation and test-execution pipelines, integrating with GitHub Actions CI/CD to synthesize regression test suites, validate execution logs, and flag behavioral drifts automatically.
- Established automated evaluation harnesses using DeepEval and synthetic golden datasets, enforcing safety guardrails, token cost budgets, and deterministic schema outputs.

AI Software Engineer • *Freelance & Enterprise Client Projects*
Jan 2023 – Sep 2024 • Hybrid / Italy

- Developed asynchronous microservices in typed Python (FastAPI, Pydantic, asyncio) serving LLM reasoning and entity classification pipelines with sub-second response times.
- Implemented semantic document ingestion pipelines parsing complex multi-format enterprise assets into chunked vector repositories with metadata filtering.
- Containerized agent services with Docker and created automated continuous integration workflows in GitHub Actions with strict type checking (mypy) and unit test coverage.
- Integrated automated observability and distributed tracing with OpenTelemetry, capturing granular token usage metrics and operational error budgets.
- Authored clean technical documentation, design specifications, and code review standards, promoting maintainable software practices across engineering teams.

Featured Autonomous Systems & Engineering Projects

AI-Powered Autonomous Code Pipeline & Test Generator • *Lead Architect*
Production Reference • 2025 – 2026

- Built an autonomous engineering pipeline that ingests ticket specifications, explores codebases via MCP tool calling, generates typed Python implementations, and creates paired pytest suites.

- Achieved a 94% pass rate on automated execution gates with automated self-healing feedback loops that parse test tracebacks and patch offending logic iteratively.

High-Precision Enterprise Knowledge RAG Engine • *Architecture & Build* Production Reference • 2024 – 2025

- Implemented a hybrid retrieval engine across 150k+ technical documents utilizing pgvector and Redis semantic caching, reducing model inference costs by 45% and boosting retrieval recall to 96%.
- Deployed automated evaluation hooks tracking faithfulness, context relevance, and answer accuracy via RAGAS.

Education, Credentials & Languages

- **Bachelor of Arts in Philosophy & Economics** • Ca' Foscari University of Venice (Coursework in Formal Logic, Quantitative Systems & Analytical Modeling)
- **Languages:** English (C1 – Full Professional Proficiency), Italian (Working Proficiency), Spanish (Basic / Rapid Learner), Hindi (Native)
- **Work Authorization:** EU Permanent Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) • Unrestricted EU right to work with no visa sponsorship required • Immediate availability for relocation to Madrid.